

23.1 A 32Gb/s/λ 256Gb/s/Fiber Half-Rate Bandpass-Filtered Clock-Forwarding DWDM Optical Link in a 3D-Stacked 7nm EIC/65nm PIC Technology

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Abstract

We present a 32Gb/s/λ dense wavelength division multiplexing bandpass-filtered clock-forwarding link featuring nine 200GHz-spaced wavelengths (8 for data and 1 for clk). Fabricated in a 3D-stacked technology with an electronic IC in a 7nm process and a photonic

IC in a 65nm SiPh process, our implementation achieves a throughput of 256Gb/s/fiber, delivering an energy efficiency of approximately 2.78pJ/b and an area efficiency of 1.33Tbps/mm²—while maintaining 0.46UI wide eye openings at <1E-11 BER.

The rapid growth of data centers, driven by increasing demand from artificial intelligence training and inference, has highlighted the limitations of traditional copper cables in terms of density, reach, and energy efficiency. Photonic links, especially dense wavelength division multiplexing (DWDM) links [1-10], offer the capability to break these barriers and expand the all-to-all connection domain to thousands of processors. DWDM links leverage multiple wavelengths, each of which transmits data at a modest rate (typically in the lower tens of gigabits per second). This strategy eliminates the need for ultra high-speed signaling (such as 100+Gb/s), improving the area and energy efficiency as well as reducing system complexity.

High-density, low-power electrical chip-to-chip (C2C) links [11-14] are a promising way to connect hosts to DWDM optical engines (OE), commonly using forward clocking (FC) due to its excellent jitter tracking bandwidth. There is a strong interest in adapting the FC scheme to DWDM links, aiming to preserve its advantages. However, DWDM links are predominantly impacted by transimpedance amplifier (TIA) thermal noise-induced jitter. Simply forwarding a clock could substantially increase the jitter power, potentially doubling it at the point of data slicing. Consequently, embedded clocking (EC) techniques [1-4] are widely adopted, which come with additional Clock and Data Recovery (CDR) challenges and exposure to jitter beyond its limited bandwidth, such as the transmitter phase-locked loop (TXPLL) induced jitter and supply-induced jitter.

We propose a half-rate FC DWDM link architecture for co-packaged optics (CPO), shown in Fig. 23.1.1, where eight wavelengths carry data and one wavelength carries a half-rate FWDCLK, which is band-pass filtered at the receiver (RX) for data recovery. A band-pass filter with a bandwidth of 1 to 2GHz rejects much of TIA thermal noise induced uncorrelated random jitter on the clock because the noise bandwidth (BW) is comparable to the TIA BW. In contrast, it tracks most of the correlated jitter on the data, such as TXPLL jitter and supply noise-induced jitter, since its power is primarily concentrated at low frequencies. This enables clock forwarding with reduced noise, and thus combines the main benefits of both schemes, as shown in Fig. 23.1.2 [15].

The transmitter (TX) is shown in Fig. 23.1.3. A ring oscillator (RO) based TXPLL under a regulated supply domain is shared among nine identical TX lanes with four clock phases distributed at half rate. Each TX lane includes a phase interpolator to adjust for the half unit interval (UI) phase shift between the FWDCLK and data to align the RX de-serializer (DES) sampling clock to the center of the eye diagram. Furthermore, this compensates for per-lane skews caused by routing mismatch and chromatic dispersion. It offers improved clock and data path delay matching over the RX side PI scheme and streamlines RX clocking by eliminating the need for quadrature phase generation and distribution. A 16-to-1 serializer is followed by a single-ended-to-differential converter (S2D), TX driver, and a broadband TX level shifter. The latter biases the ring cathode at high voltage, nominally 1.5×VDDA, and the anode at low voltage, nominally 0.5×VDDA. Compared with a conventional bias-T level shifter, this TX broadband level shifter does not have a lower cutoff frequency, posing no restrictions on data encoding; also, it supports microring modulator (MRM) leakage current up to 300μA [16]. By setting the input parallel data pattern as 0101, any TX lane can transmit the FWDCLK.

The RX is shown in Fig. 23.1.4. An RX phase-locked loop (RXPLL) generates the reference voltage for the TIA regulator and RX regulator, ensuring the supplies for the TIA and other RX circuits track speed and process-voltage-temperature (PVT) variations [11]. This RXPLL acts as a reference generator as its clocks are not distributed by default. At the expense of the RXPLL and regulator overhead, the bandwidth of the TIA remains relatively constant across PVT corners, with sensitivity only to resistor variation. All TIAs share the same supply domain, isolated from the rest of the RX circuits to minimize supply noise over the TIA due to its higher sensitivity. The inverter-based TIA consists of a shunt-feedback amplifier followed by two Cherry-Hooper stages. An injection-locked oscillator (ILO) stage is inserted between the outputs of the 4th and 2nd inverters with tunable natural frequency, which is ON only if it receives FWDCLK. Together with the clock distribution ILO (see below), this structure forms a bandpass filter described above that significantly suppresses the wideband uncorrelated jitter while tracking low-frequency correlated jitter. Following the TIA, two delay paths are employed: one directs the buffered output to the 1:16 DES upon

data reception, while the other routes the signal to the RX clock distribution upon FWDCLK reception.

For DWDM systems, a fixed TX-RX ring mapping (e.g., TX ring_n to RX ring_n) can lead to high heater power consumption in the presence of large wafer-to-wafer ring resonance frequency variation [8]. To minimize the heater power, the system should support a flexible TX-RX ring mapping, which causes ambiguity in RX clock distribution as the FWDCLK can be received by any RX ring. Instead of a 9-to-1 mux at the source of RX clock distribution, which limits bandwidth and introduces an extra hurdle for clock to data matching, an ILO structure is embedded into the RX clock distribution. As shown in Fig. 23.1.4 (bottom), one lane drives the distributed net and others oscillate at the operation frequency or stay OFF. This allows the FWDCLK to be sourced from any lane and provides additional bandpass filtering. To reduce parasitic capacitance, the RX clock distribution is split into two groups that are both connected to the 2:1 input clock mux in all DESs, with only one group active during normal operation. Besides the jitter filtering benefit, this ILO-based RX clock distribution scheme also improves the resilience of the system to potential laser reliability issues and provides the possibility to assign desired wavelengths to FWDCLK for performance optimization. Compared with the quarter-rate FWDCLK scheme [8], this approach avoids multi-phase generation at the RX side, as such circuitry usually leads to unwanted mismatch between the data and clock paths.

Both TX and RX rings are tuned thermally to lock to the target wavelength with on-chip mixed-signal loops. The TX thermal tuning (TT) loop locks to a dialed-in drop port photodetector (PD) current level, determined by a pre-characterization step prior to transmitting live data. The RX thermal tuning loop taps the TIA DC-loop to maximize the RX PD DC current. As shown in Fig. 23.1.3 bottom (right), for either loop, the current reading is first digitized by an 8b temperature-insensitive switched cap-based combined integrator/successive approximation register analog-to-digital converter (SAR ADC). It is then processed digitally (either with proportional-integral control [TX] or a peak-finding algorithm [RX]) to form a pulse density modulated (PDM) signal to drive a metal heater co-integrated with the ring for maximum energy efficiency.

This architecture employs a 7nm electronic integrated circuit (EIC) stacked atop a 65nm silicon photonics (SiPh) photonic integrated circuit (PIC), interconnected through hybrid bonding technology to minimize interface parasitics and maximize receiver sensitivity, as shown in Fig. 23.1.7. Each TX PHY includes nine data/FWDCLK lanes, one TXPLL lane, one regulator lane for the TXPLL, and an extra lane for decoupling capacitors. Each RX PHY has nine data/FWDCLK lanes, one RXPLL lane, and two regulator lanes for the TIA and other RX circuits respectively. Each lane measures 80.94×80.94 μm² with an extra 18.012mm-wide routing corridor for parallel data, control, and status bits. The EIC is connected to the PIC via 9μm-pitch hybrid bonding pads, minimizing parasitics by shortening the routing distance from the TX driver to the TX ring and RX PD to the RX TIA. This is achieved through positioning each ring with PD directly below the corresponding electrical lane. The PIC features multiple ~5μm radius rings along the waveguide for DWDM experiments, with grating couplers used for fiber connections. The TX ring modulator quality factor Q is ~4.5K, and RX filter Q is 4K, selected using an in-house link modeling tool, and validated by device test chip measurements. We place the RX rings in radius-descending order to help partially cancel optical crosstalk [17].

A PIC TX bus spectrum sweep is shown in Fig. 23.1.5 top (left). All 9 rings are spread around 1310nm with about 200GHz spacing. The TT loop response to the input laser wavelength steps is shown in Fig. 23.1.5 top (right). The heater code follows the laser wavelength shift properly, confirming the functionality of the TT loop. Generated by a tunable laser source, a single wavelength is modulated by one TX ring with on-die generated PRBS7 pattern. The eye diagram of the modulated NRZ signal is shown in Fig. 23.1.5 bottom (left), indicating a robust TX. With one data lane and one FWDCLK lane on, the RX sensitivity is measured at -16.0dBm OMA at BER<1E-11 level using a PRBS7 pattern. A multi-wavelength laser source is used for the link measurement, with all TX lanes active at 32Gb/s using PRBS31 patterns (varied seeds), and the FWDCLK lane (TX2) set to 16GHz. Thermal loops lock all rings to their corresponding wavelengths. One RX ring (RX6) receives the FWDCLK, filters the jitter with the TIA with ILO enabled and sends it to all the lanes with the RX clock

distribution ILO ON. The bathtub of the bit error rate (BER) is captured by sweeping the TX PI code and querying RX PRBS checker for each lane. As shown in Fig. 23.1.5 bottom (right), with all lanes simultaneously ON, the aggregate eye opening is 0.46 UI wide at $\text{BER} < 1\text{E-}11$ level. This work is summarized together with recent related publications in Fig. 23.1.6. Compared with the state-of-the-art DWDM links, it achieves $\sim 6\times$ higher shoreline and $\sim 20\times$ higher area density, important for future CPO applications in AI datacenters. Assuming 10% fiber-coupled laser wall-plug efficiency [18,19], laser energy efficiency is 0.76pJ/b. Combined with 0.76pJ/b energy for TX+RX ring heaters and 1.26pJ/b energy for TX, RX, clocking and TT circuits, it achieves 2.78pJ/b overall energy efficiency.

Acknowledgement:

The authors extend their sincere gratitude to NVIDIA's MSD team for their support with EIC/PIC design, layout, and testing; to the ATG team for CAD and tapeout assistance; to the Israel CPO team for their contributions to packaging, FAU, and substrate design; and to the NBU team for their contributions to PCB design.

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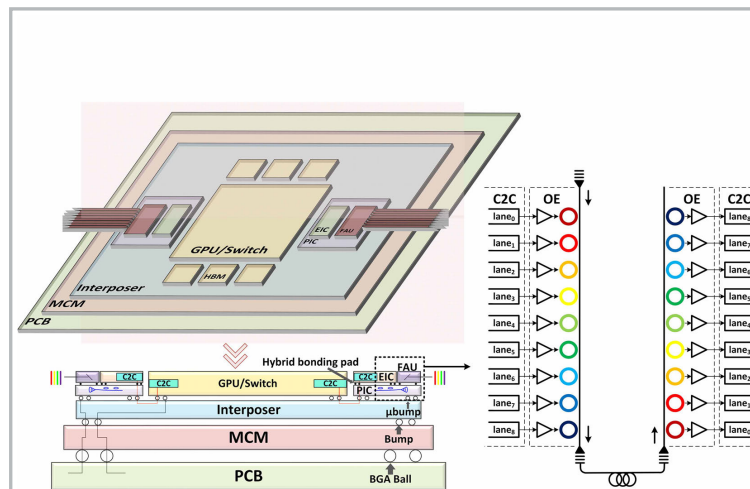


Figure 23.1.1: The proposed clock-forwarding DWDM links for CPO. One of the nine wavelengths, each spaced at 200GHz, is allocated to the FWDCLK, while the remaining eight are designated for data transmission.

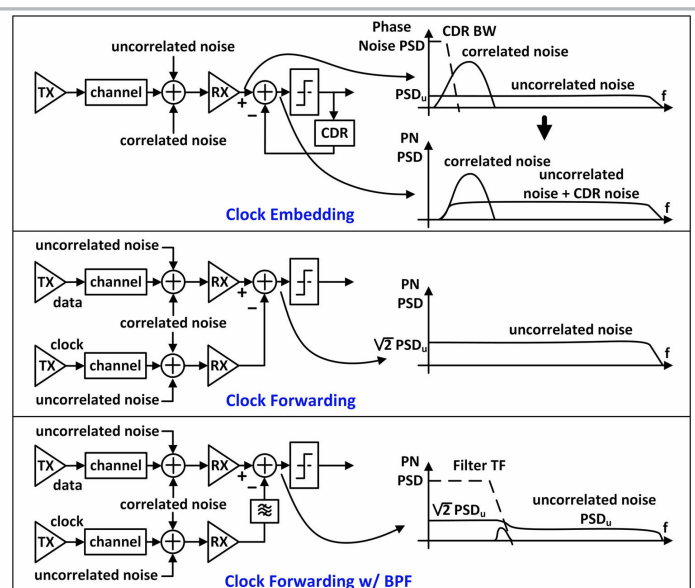


Figure 23.1.2: Phase noise comparison between the clock embedding (top), clock forwarding (middle), and clock forwarding with band-pass filtering (bottom).

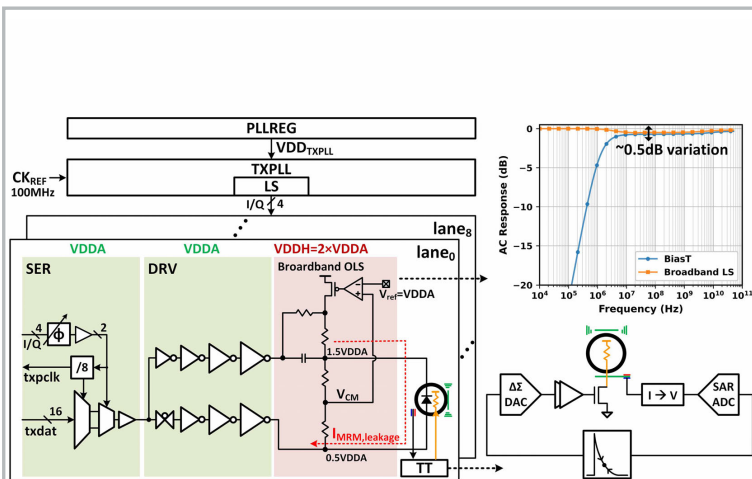


Figure 23.1.3: Left: TX PHY architecture; right: (top) the AC transfer functions of the proposed broadband level shifter vs a typical bias-T based output level shifter; (bottom) thermal tuning loop architecture.

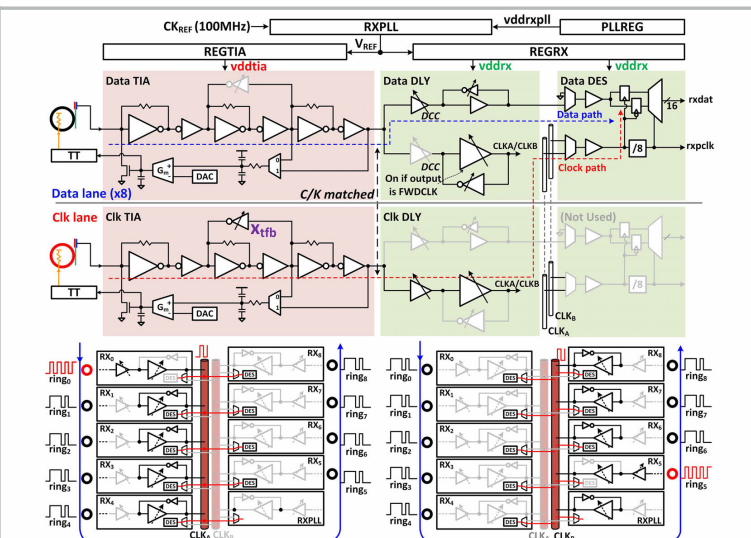


Figure 23.1.4: Top: RX PHY architecture. Feedback inverter x_{10} ON for FWDCLK reception. Bottom: (left) RX clock distribution when ring₅ receives the FWDCLK. Ring₀ path drives CLK_A while ring₁₋₄ injection locking; (right) when Ring₅ receives the FWDCLK.

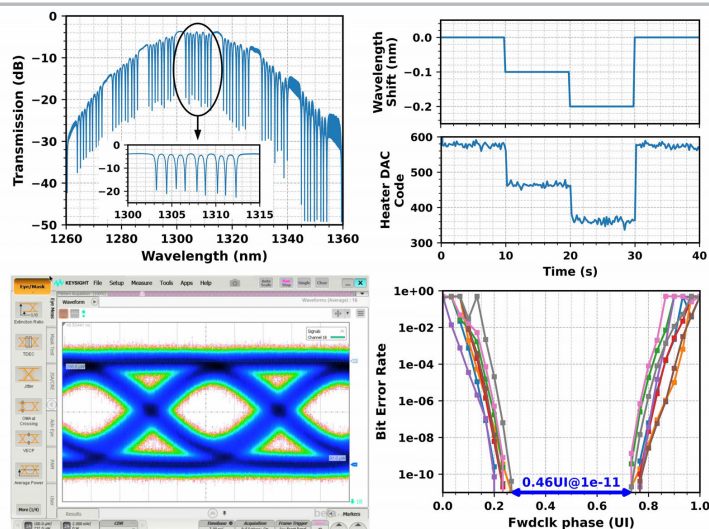


Figure 23.1.5: Top: (left) Frequency sweep of TX bus; (right) Laser wavelength shift and TX thermal tuning loop response. Bottom: (left) DCA captured TX output eye diagram for one λ at 32 Gb/s, PRBS7 pattern; (right) Bathtub curves with all lanes on simultaneously, PRBS31 pattern, TX2-RX6 pair for FWDCLK.

Process	EIC	This work	VLSI'22 [1]	VLSI'23 [9]	ISSCC'23 [3]
		7nm FinFET 65nm SOI SiPh	45nm SOI CMOS	28nm CMOS 300nm SOI SiPh	7nm CMOS 45nm SOI SiPh
Integration		Hybrid	Monolithic	Hybrid	Hybrid
Architecture		FC	EC	EC	EC
Number of wavelengths		9	8	8	7
Per- λ Data rate <Gbps>		32	16	32	50
Per-Fiber Data rate <Gbps>		256	128	256	350
TX+RX Lane size < μm^2 >		0.013	-	~1	0.031 (RX only)
EIC shoreline density ^(a) <Tbps/mm>		0.8	~0.14	~0.064	-
EIC area density ^(b) <Tbps/mm ² >		1.33	~0.064	~0.032	-
RX sensitivity OMA <dBm>		-16.0 ^(c)	-	-10	-10.1 to -11.4
Energy efficiency <pJ/b>	BER	<1E-11	<6.9E-12	<1E-12	<1E-12
	Laser	0.76 ^(d)	-	-	-
	Thermal	0.76 ^(e) 0.49 ^(f)	<0.6	-	-
	TRX + Clock + TT circuits	1.26	4.9-5	3.8 (RX only)	0.96 (RX only)
	Total	2.78 2.51	-	-	-

(a) Bi-directional, 1-PHY deep. (b) Bi-directional. (c) Measured with one data lane and one FWDCLK ON, PRBS7 pattern. (d) Assuming 10% fiber-coupled laser WPE. (e) Measured with the bench laser. (f) Estimated based on the measured heater efficiency, covering 3 σ of ring and modeled laser variation.

Figure 23.1.6: Comparison with state-of-the-art DWDM links of NRZ modulation.

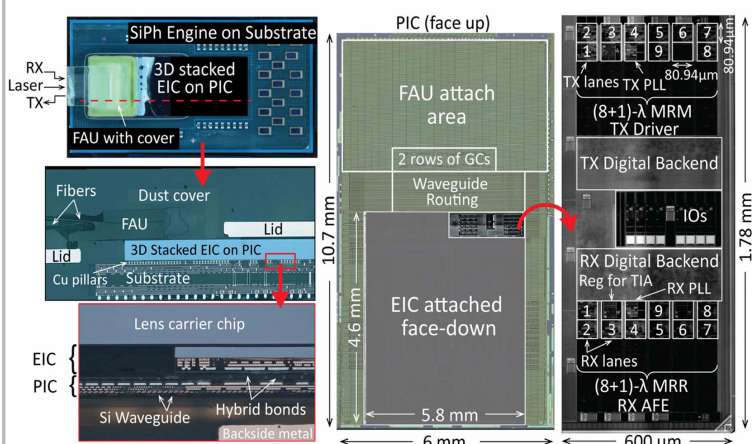


Figure 23.1.7: Optical engine on substrate, assembly cross-section, and die micrograph of EIC and PIC chips.